Chapter IV

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Functions and Continuity

1 Introductions and Motivation

For around 200 years, the term function simply meant analytical expression or formula. That is, expressions consisting entirely of sums,products or square roots of polynomials, sines, cosines, exponentials etc. By their very nature, functions were always smooth and could not contain 'jumps' or gaps in their respective graphs. Further, due to the huge successes of power series, namely series of the form

$$\sum_{k=0}^{\infty} a_k x^k,$$

and being able to write complicated functions as power series, it was the belief to many mathematicians of the time, in particular Euler and Lagrange, that all functions could be written as a power series. Today, we call this collection of functions *analytic functions*. Under this understanding of function, and belief that all functions could be written as power series, it would mean that every function could be completely determined by its behaviour at a single point. It was the work of Fourier in 1807, and some early work by D'Alembert and the dynamics of strings, that seemed to contradict this understanding of function. In particular, Fourier was able to show that by studying the propagation of heat in bodies that

$$1 = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{2k-1} \cos\left(\frac{(2k-1)\pi x}{2}\right)$$

for $-1 < x < 1$. That is, this series, aptly now called a *Fourier series*, was equal to the constant function 1 when x was between -1 and 1. If this equation were true, it fundamentally conflicted with the understanding of the term *function*, as understood at the time. This can be seen by shifting the Fourier series by 2, so the transformed function is now given by

$$-1 = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{2k-1} \cos\left(\frac{(2k-1)\pi(x+2)}{2}\right).$$

But this seems to contradict the idea that if the value of a function is known at a particular point, then the entire function can be determined. If this were true, then surely the function should be constant for all $x \in \mathbb{R}$ and not bouncing between 1 and -1. Further, at the points $x = 1$ and $x = -1$, the series converges to 0. This was by no means a function according to the understanding of function at the time. This led people to go back and broaden the definition of function so as to encompass examples such as the one Fourier developed. Additionally, with this broader definition of function, it also meant that many notions in calculus that were taken as given, or trivial, had to be developed on a much more rigorous footing. We will begin to explore these in the next two chapters.

2 Functions

The modern definition of function is owed to Dirichlet. Today, by function we simply mean the assignment of one set of values to another. More concretely, its definition is given as follows.

Definition 2.1: Function

A ***function*** $f : S \to T$, with S and T sets, associates to each element $x \in S$ a unique element $f(x) \in T$. The set S is called the ***domain*** of f and the set T is called the ***co-domain*** of f.

Since this course is entitled *real analysis*, we will exclusively be looking at functions with co-domains in $\mathbb{R}$, or subsets of $\mathbb{R}$. We will see that the algebraic and ordering structures we defined on $\mathbb{R}$, as well as its completeness property, will allow these class of functions to exhibit interesting properties that otherwise would not exist for more general co-domains. This being said, we have a general definition for a function with co-domain in $\mathbb{R}$.

Definition 2.2: Real Function

A ***real function*** is a function with $\mathbb{R}$ as its co-domain.

Now, the astute reader may have noticed that sequences are simple examples of real functions in which the domain of the function is always taken to be $\mathbb{N}$. In the case of sequences, it was easy to see when two were equal if their respective formulas for the n-th terms a_n were equal. However, this is not the case for more general real functions like we are considering here. This is because there are two variables at play. Namely, the domain of the function, and the formula $f(x)$ the defines the assignment of each value in the domain. This is outlined explicitly in some of the following examples:

Example 2.3: Real Functions

(i) Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ and $g : (1, \infty) \to \mathbb{R}$ given by

$$f(x) = \sin\left(\frac{1}{x}\right) \text{ and } g(x) = \sin\left(\frac{1}{x}\right).$$

Note that $f(x) \neq g(x)$ since the domains are different.

(ii) Functions can also be defined ***piecewise***, For example, the function $h : \mathbb{R} \to \mathbb{R}$ given by

$$h(x) = \begin{cases} \sin(x) : x \geq 0 \\ \cos(x) : x < 0 \end{cases}$$

(iii) Let $d : \mathbb{R} \to \mathbb{R}$ given by

$$d(x) = \begin{cases} 1 : x \in \mathbb{Q} \\ 0 : x \notin \mathbb{Q} \end{cases}$$

This function is called the ***Dirichlet function.***

As already mentioned, sequences are example of real functions. One important feature of them though was that their domains were discrete. That is, there does not always exist a number between any two elements of a discrete set, like $\mathbb{N}$. In this chapter, we are going to be interested in real functions whose domains are intervals of $\mathbb{R}$. The reason for this is that $\mathbb{R}$ was constructed to be complete. Loosely speaking, this meant that there were no 'gaps' in the number system. We would like to see how different functions interact with these sets that contain no gaps. What type of functions preserves this property and which do not? To this end, we start with a definition for the various intervals of $\mathbb{R}$ that could appear.

Definition 2.4: Real Intervals

An ***interval*** I is a subset of $\mathbb{R}$ that contains all real numbers between any two numbers within the subset. They are either of the form:

(i) $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$ or $(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$. These are referred to as ***clopen*** or ***half-open*** intervals.

(ii) $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$ or $[a, \infty) = \{x \in \mathbb{R} : x \geq a\}$. These are also referred to as ***clopen*** or ***half-open*** intervals.

(iii) $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$. These are also referred to as ***closed*** intervals.

(iv) $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ or $(-\infty, b) = \{x \in \mathbb{R} : x < b\}$ or $(a, \infty) = \{x \in \mathbb{R} : x > a\}$. These are also referred to as ***open*** intervals.

In each case, it is taken that $a < b$ with $a, b \in \mathbb{R}$.

2.1 New Functions From Old

With this new definition of function in hand, we are now able to consider the construction of new functions from old ones. This is similar to the old fashioned understanding of function in which now functions could be constructed by taking sums or products of known ones. We would expect this to be true with our broader definition of function and indeed this is the case via the following definition.

Definition 2.5: Algebra on Real Functions

Let $f : S_1 \to \mathbb{R}$ and $g : S_2 \to \mathbb{R}$. Define $S := S_1 \cap S_2$. Then, the functions $f + g, fg, f/g$ and cf are defined pointwise by:

(i) $(f + g)(x) = f(x) + g(x)$ for all $x \in S$

(ii) $(fg)(x) = f(x)g(x)$ for all $x \in S$

(iii) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ providing $g(x) \neq 0$ for all $x \in S$.

(iv) $(cf)(x) = cf(x)$ for all $c \in \mathbb{R}$ and $x \in S$.

Additionally, the function $|f| : \mathbb{S}_1 \to \mathbb{R}$ is defined as

$$|f|(x) = |f(x)|$$

for all $x \in S_1$.

Remark 2.6: Function Spaces

An interesting/important comment here is that despite using the same notation for addition, namely $+$, the symbol $+$ between f and g is fundamentally different to the one between $f(x)$ and $g(x)$. The first one that defines $f + g$ is an algebraic structure that takes in two functions and produces a new function. The second $+$, that defines $f(x) + g(x)$,is the addition between two real numbers. So whilst the same notation is used for both, they act on different objects. This is often seen throughout all of mathematics, and in part is a laziness of mathematicians. It is normally 'obvious' from the context that they mean different structures, even though the notation is the same. You will have most likely seen this throughout linear algebra and in the more general definition of vector spaces. Finally, this is very similar to how $+$ was used for series. Despite using this notation, it did not mean $+$ on $\mathbb{R}$. In fact, for series it isn't even an addition at all, just an abuse of notation.

One final operation to consider on functions is that of composition. That is, two functions can be sewn together, subject to some conditions on their domains/co-domains, to produce a new function. This is defined as follows.

Definition 2.7: Composition of Functions

Let $f : S \to D$ and $g : D \to T$. Then the ***composition*** $h : S \to T$ of g and f is given by

$$h(x) = g(f(x))$$

and is denoted by $h = g \circ f$.

2.2 Bounded Functions

Much like the section on sequences, analysing the whole space of real functions from the start is a tumultuous task. Instead, it makes sense to start by studying a smaller class of functions, namely the bounded functions. Importantly, we will see that this class of functions will always appear subject to certain conditions on the domain of the function. In turn, this will lead to the development of perhaps one of the most important theorems in calculus, namely the *intermediate value theorem.* Now we are properly motivated, the definition of a bounded function is given as follows.

Definition 2.8: Bounded Functions

A real function $f : S \to \mathbb{R}$ is:

(i) ***bounded above*** if there exists $K_U \in \mathbb{R}$ such that

$$f(x) \leq K_U$$

for all $x \in S$. Then number K_U is called an ***upper bound*** for f.

(ii) ***bounded below*** if there exists $K_L \in \mathbb{R}$ such that

$$f(x) \geq K_L$$

for all $x \in S$. Then number K_L is called an ***lower bound*** for f.

(iii) ***bounded*** if it is bounded above and below.

A real function that is not bounded is said to be ***unbounded.***

As already discussed, two functions are not necessarily equal if their formulas are the same. This is due to the fact that the domains of the functions could be wildly different. So just as it is the case for equality of functions, boundedness of functions is not determined by the formula alone, but instead the combination of the formula and the domain of the functions. This is outlined in some of the following examples.

Example 2.9: Bounded/Unbounded Functions

(i) Let $f : [1, \infty) \to \mathbb{R}$ with $f(x) = 1/x$. Then this function is bounded above by 1 and below by 0. Note that the new function $g : (0, 1] \to \mathbb{R}$ given by $g(x) = 1/x$ has the same formula as f but is not the same. Additionally, g is bounded below by 1 but is now not bounded from above.

(ii) Let $f : [0, 3] \to \mathbb{R}$ given by $f(x) = x^2$. This function is bounded above by 9 and below by 0.

Finally, mirroring the sequences chapter, the definition for a bounded function can be recast into a bound on the absolute value of the function. This is captured by the following theorem.

Theorem 2.10: Bounded Function

Let $f : S \to \mathbb{R}$. Then f is bounded if and only if there exists a $K \in \mathbb{R}$ such that

$$|f(x)| < K$$

for all $x \in S$.

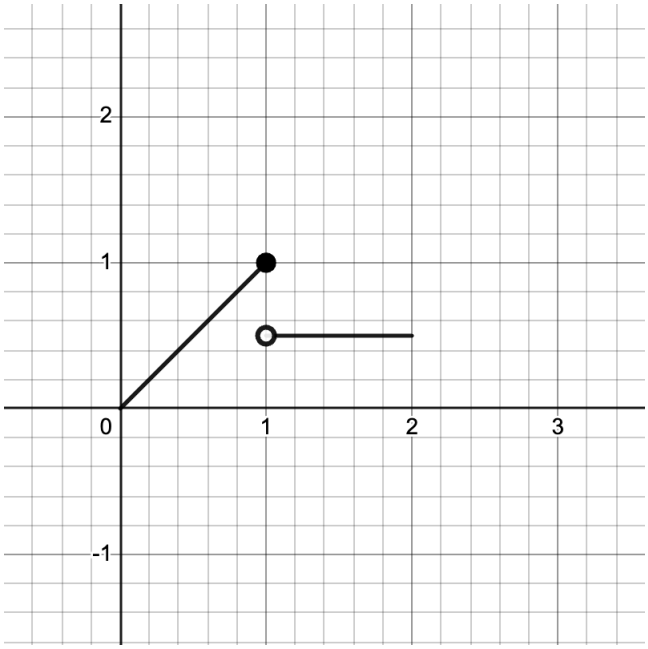
Proof. Exercise □

3 Continuous Functions

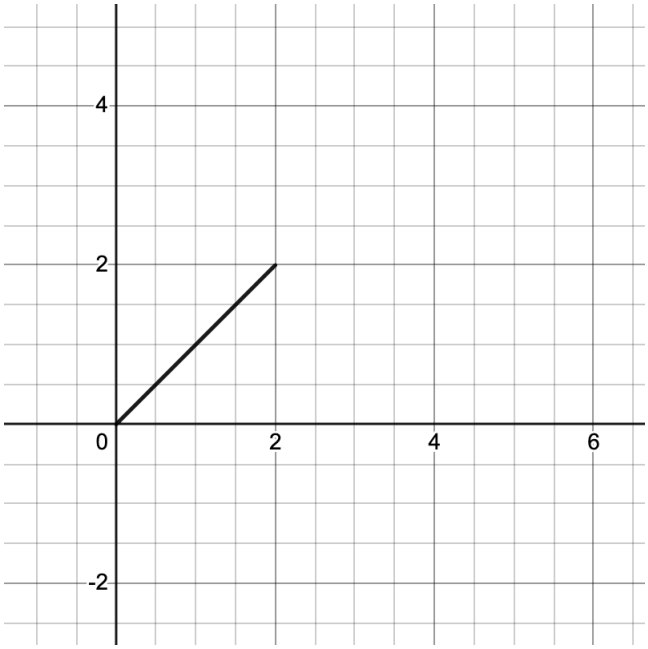
Unlike sequences, real functions, with domains as intervals in $\mathbb{R}$, have special properties emerging from the fact that $\mathbb{R}$ is complete. One property of particular interest is the nature of graphs of real functions. The old fashioned understanding of function meant that they were inherently smooth to mean that their graphs have no jumps or gaps within them. Due to this, there was no need to ever explore such distinctions between functions with gaps and ones without, since they were simply not thought of as functions, or more accurately, they were functions that were not worth studying. But, with our new broader definition of function, more exotic graphs can be constructed. For example, consider the function $f : [0, 2] \to \mathbb{R}$ given by

$$f(x) = \begin{cases} x & : x \in [0, 1] \\ 1/2 & : x \in (1, 2] \end{cases}$$

The graph of this function looks like the following:



At the point $x = 1$, the function has a gap, or jump. However, a function such as $g : [0, 2] \to \mathbb{R}$ given by $g(x) = x$, has not gaps in its graph as can be seen below:



Clearly these two functions are somehow different in their character based on their graphical representations. That is, one has gaps and the other does not. Perhaps then, we should classify all real functions according to whether they exhibit this characteristic or not. This is precisely what mathematicians defined as continuity, namely whether the graph of a function has no gaps or not. It was said that a function was continuous if its graph was unbroken. This definition remained for much of the 19th century and was even published in the core calculus text of the time up until 1858. In this text, continuity was defined as:

"By the law of continuity is meant that which is observed in the description of lines of motion, and according to which the consecutive points of the same line succeed each other without interval"

The point of this discussion is that this intuitive understanding of continuity was taken to be the case for mathematicians for several centuries and the more modern understanding of continuity was by no means a simple observation. Why then is this definition not sufficient? Well, firstly, it is useless for proving any results about continuous functions analytically or algebraically. So, perhaps by making this definition rigorous, or analytic, this solves our problem. Unfortunately, this is not the case, despite many textbooks claiming this. The definition of continuity does not directly result by making such intuitive notions of continuity rigorous. The reason for this is that the rigorous version of this definition does not exhibit the properties expected, or desired, of continuous functions. To see this, let's write down the analytic definition for functions that exhibit this intuitive notion of continuity.

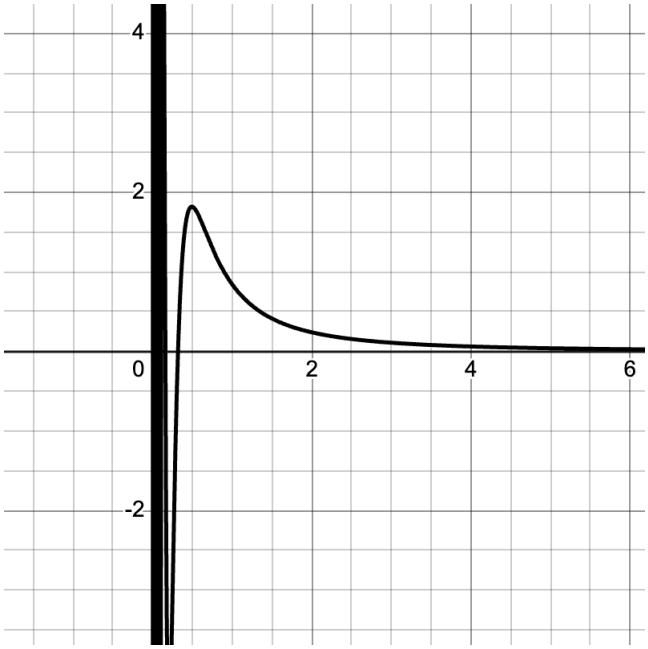
Definition 3.1: Darboux function

Let $f : D \to \mathbb{R}$. Then the function is called ***Darboux*** if for every $[a, b] \subset D$ and $\alpha \in \mathbb{R}$, with $f(a) < \alpha < f(b)$ or $f(b) < \alpha < f(a)$, there exists a $c \in (a, b)$ with $f(c) = \alpha$.

This definition simply means that if a function satisfies the properties outlined, then its graph has no gaps. That is, a Darboux function is one with no gaps. Already by its name it hints that this is not the definition of continuity we are working towards. Why is this the case though? Surely if this is the rigorous notion of a function to have no gaps in its graph, and that is our intuitive understanding of continuity, then this definition suffices. Well, there are a couple of problems with this definition. Firstly, a function can exhibit this property but could be unbounded on a closed interval $[a, b]$. For example, consider the real function $f : [0, 1] \to \mathbb{R}$ given by

$$f(x) = \begin{cases} x^{-1} \sin(1/x) & : x \in (0, 1] \\ 0 & : x = 0 \end{cases}$$

The graph of this function looks like:



It can be seen then that this indeed has no gaps and has the properties to make it a Darboux function. Yet, it is unbounded on the interval $[0, 1]$. We would like it to be the case that whatever our definition of continuity, that continuous functions on closed intervals $[a, b]$ are also bounded. This is not the case with this definition. The second issue is that the sum of two Darboux functions may not be a Darboux function. That is, adding two functions that contain no gaps in their graphs, according Definition 2.5, could yield a new function that contains gaps. This can be seen via the functions $g : [0, 1] \to \mathbb{R}$ and $h : [0, 1] \to \mathbb{R}$ given by

$$g(x) = \begin{cases} \sin^2(1/x) & : x \in (0, 1] \\ 0 & : x = 0 \end{cases} \quad \text{and} \quad h(x) = \begin{cases} \cos^2(1/x) & : x \in (0, 1] \\ 0 & : x = 0 \end{cases}.$$

Then $g + h : [0, 1] \to \mathbb{R}$ is given by

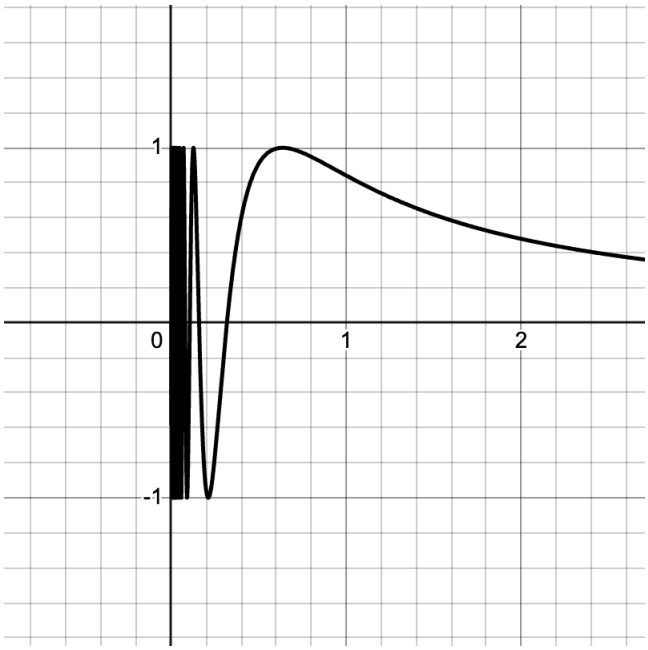
$$(g + h)(x) = \begin{cases} 1 & : x \in [0, 1) \\ 0 & : x = 0 \end{cases}.$$

The function $g + h$ clearly contain gaps in the interval $[0, 1]$ whilst g and h so not. This is not a drastic problem, but again, we would ideally like our definition of

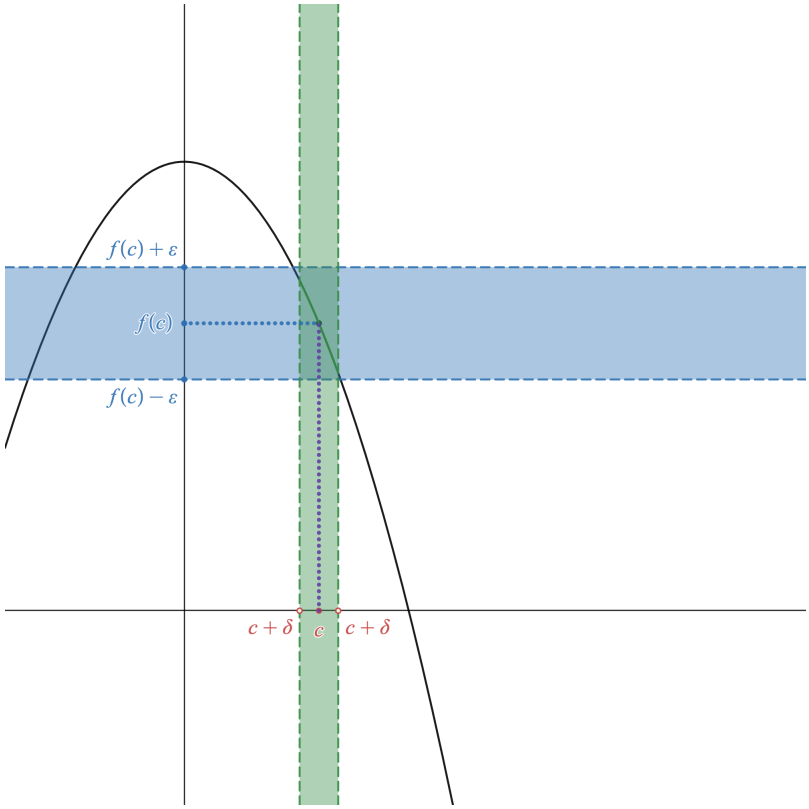
continuity to be preserved under function addition.
The last problem is perhaps the the worst property, and the one that in resolving will play a core role in the correct definition of continuity. This is, small errors in the input of the function can lead to large errors in the output of the function, even if the function contains no gaps. A standard example of this is the *topological sine curve*. This is the function $f : [0, 1] \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \sin(1/x) & : x \in (0, 1] \\ 0 & : x = 0 \end{cases}$$

The graph of this function looks like the following:



This function satisfies the property of having no gaps, but, any error in the input around 0 yields an output that could be any number between -1 and 1. That is, no matter how close we get to the point 0, the function $f(x)$ does not get closer to one specific point. It turns out by resolving this problem, and considering functions that do no exhibit this property, this not only resolves the first and second problems but also gives us a sensible definition for continuity. What does it mean then for this third property to not hold? Well, we require that as a region around an input value, say c, of a function is squeezed, that the region around the output value $f(c)$ can become arbitrarily narrow. More precisely, we want to be able to force $f(x)$ to be arbitrarily close to $f(c)$ by controlling the distance between x and c. This can be pictorially realised as follows:



With this picture in mind, we finally arrive at our definition of continuity.

Definition 3.2: Topological Continuity

A real function $f : D \to \mathbb{R}$ is ***continuous at the point*** $c \in D$ if or every ε-neighbourhood $V_\varepsilon(f(c))$ of $f(c)$, there exists a δ-neighbourhood $V_\delta(c)$ of c such that $f(x) \in V_\varepsilon(f(c))$ whenever $x \in V_\delta(c)$ and $x \in D$.

Now, whilst this definition is motivated from the picture, much like the definition for convergent sequences, it is the case that it isn't of much use for proving specific statements regarding continuous functions. Instead, we must unravel the components of this definition so we are able to utilise it to prove specific theorems as well as the continuity of specific functions at particular points. Unpicking this definition using the definition of neighbourhoods on $\mathbb{R}$ gives the following.

Definition 3.3: Metric Continuity

A real function $f : D \to \mathbb{R}$ is ***continuous a point*** $c \in D$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - f(c)| < \varepsilon$ whenever $|x - c| < \delta$ and $x \in D$.

This definition is the more common one found in real analysis or calculus texts and is the one we will utilise to prove the continuity of specific functions at certain points. It is emphasised though that both definitions are correct, with the first being easier to construct from a picture. Without further a due, let's prove that some fairly tepid functions are indeed continuous at specific points.

Example 3.4: Continuous Function

Let $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 10x$. Then f is continuous at the point $c = 2$.

Scratch Work. We ideally want to construct an inequality of the form

$$|10x - 20| < k|x - 2|,$$

for some $k \in \mathbb{R}$. This is because we are able to choose a $\delta > 0$ so that $|x-2| < \delta$. That is, we can make the bound on $|x-2|$ as small as necessary to ensure that for very $\varepsilon > 0$ that $|10x - 20| < \varepsilon$. With this idea in mind, some simple algebra shows that

$$|10x - 20| = 10|x - 2|.$$

Now, if it is chosen that for each $\varepsilon > 0$ that $\delta = \varepsilon/10$, we would have our desired inequality. We are now able to form the proof.

Proof. For every $\varepsilon > 0$, define $\delta = \varepsilon/10$. It follows than that if $|x - 2| < \delta$, we have

$$|10x - 20| = 10|x - 2| < 10\delta < \varepsilon.$$

□

Note that much like the case for proving the convergence of sequences directly from the ε-definition, proving the continuity of a given function at a specific point could be exceedingly difficult depending on the function given. For example, proving that $f : (0, \pi/2) \to \mathbb{R}$ given by $f(x) = \sqrt{\tan(\sec^2(x))}$ is continuous at the point $c = 1$, directly from the ε-δ definition of continuity, would be an awful thing to prove. The purpose of this discussion is to emphasise that the proofs of continuity of functions at points can differ wildly and therefore there is no concrete algorithm that can be followed to produce such proofs. To demonstrate this point, here is another proof of continuity at a point, but for a slightly more complicated function.

Example 3.5: Continuous Function

Let $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$. Then f is continuous at the point $c = 3$.

Scratch Work. Like before, we want to relate the expression $|x^2 - 9|$ to $|x - 3|$. This is because we are able to place bounds on $|x - 3|$ which in turn should allow us to place bounds on $|x^2 - 9|$. With this in mind, a short calculation shows that

$$|x^2 - 9| = |x + 3||x - 3|.$$

This is trickery than the previous example though since we have a function multiplying $|x - 3|$ and not a constant. Since δ is a fixed number that can only depend on ε, we cannot make it a function of x. What we want to do then is restrict the region around the point c, in this case $c = 3$, so that we

have a bound on $|x+3|$. To this end, suppose

$$|x-3|<1.$$

It follows then that $|x+3|<7$ and hence $|x^2-9|<7|x-3|$. All we need to do then is ensure two inequalities are true. Namely that

$$|x^2-9|<\varepsilon \text{ and } |x+3|<7.$$

With this in mind, let's construct our proof.

Proof. For every $\varepsilon>0$, let $\delta=\min(1,\varepsilon/7)$. Then, if $|x-3|<\delta\leq 1$ it follows that

$$\begin{aligned}|x^2-9| &= |x+3||x-3|\\ &< \delta|x+3|\\ &\leq \frac{\varepsilon}{7}|x+3|\\ &< \varepsilon.\end{aligned}$$

□

Obviously it is rare for functions to just be continuous at a single point. Often, functions are continuous over their entire domains, or for some subsets of their domains. Given this, we have an additional definition to encompass these types of continuous functions.

Definition 3.6: Continuous on Interval

Let $f:D\to\mathbb{R}$. Then the function f is said to be ***continuous on the interval*** I if it is continuous at each point $c\in I\subset D$, it

It is worth commenting that despite this definition, in order to prove continuity over an interval $I\subset D$ it is still necessary to show it is continuous at each point $c\in I$.

4 Properties of Continuous Functions

Recall that one of the problems with defining functions with no gaps as continuous functions was that these functions were not closed under addition. With our definition of continuity, we are able to show that not only are continuous functions at a point closed under addition, but also under scalar multiplication, multiplication and division. All of these properties are proven in the following theorem.

Theorem 4.1: Algebraic Properties of Continuous Functions

Let $f:D\to\mathbb{R}$ and $g:D\to\mathbb{R}$ both be continuous at the point $c\in D$. Then it follows that:

(i) $f+g:D\to\mathbb{R}$ is continuous at $c\in D$.

(ii) $fg:D\to\mathbb{R}$ is continuous at $c\in D$.

(iii) $f/g : D \to \mathbb{R}$ is continuous at $c \in D$, providing $g(x) \neq 0$ for all $x \in D$.

(iv) $\lambda f : D \to \mathbb{R}$ is continuous at $c \in D$ for all $\lambda \in \mathbb{R}$.

Proof. (i) For every $\varepsilon > 0$, there exists a $\delta_1 > 0$ such that

$$|f(x) - f(c)| < \frac{\varepsilon}{2}.$$

whenever $x \in D$ and $|x - c| < \delta_1$. Similarly, there exists a $\delta_2 > 0$ such that

$$|g(x) - g(c)| < \frac{\varepsilon}{2}$$

whenever $x \in D$ and $|x-c| < \delta_2$. Let $\delta = \min(\delta_1, \delta_2)$. Then, for every $\varepsilon > 0$, if $|x - c| < \delta$ and $x \in D$, it follows that

$$\begin{aligned} |f(x) + g(x) - f(c) - g(c)| &\leq |f(x) - f(c)| + |g(x) - g(c)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

□

Proof. (ii) Since, f is continuous at c, there exists a $\delta_1 > 0$ such that

$$|f(x)| < |f(c)| + 1$$

whenever $x \in D$ and $|x - c| < \delta_1$. Similarly, for every $\varepsilon > 0$, there exists a $\delta_2 > 0$ such that

$$|f(x) - f(c)| < \frac{\varepsilon}{2(|g(c)| + 1)}$$

whenever $x \in D$ and $|x - c| < \delta_2$. Finally, there exists a $\delta_3 > 0$ such that

$$|g(x) - g(c)| < \frac{\varepsilon}{2(|f(c)| + 1)}$$

whenever $x \in D$ and $|x - c| < \delta_3$. Let $\delta = \min(\delta_1, \delta_2, \delta_3)$. Then, for every $\varepsilon > 0$, if $x \in D$ and $|x - c| < \delta$, if follows that

$$\begin{aligned} |f(x)g(x) - f(c)g(c)| &= |f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c)| \\ &\leq |f(x)||g(x) - g(c)| + |g(c)||f(x) - f(c)| \\ &< (|f(c)| + 1)\frac{\varepsilon}{2(|f(c)| + 1)} + |g(c)|\frac{\varepsilon}{2(|g(c)| + 1)} \\ &< \varepsilon/2 + \varepsilon/2 \\ &= \varepsilon. \end{aligned}$$

□

Proof. It suffices to prove that $1/g$ is continuous at the point c. Since g is continuous at the point $c \in D$, it follows that there exists a $\delta_1 > 0$ such that

$$\frac{|g(c)|}{2} < |g(x)|,$$

whenever $x \in D$ and $|x - c| < \delta_1$. Similarly, for every $\varepsilon > 0$, there exists a $\delta_2 > 0$ such that

$$|g(x) - g(c)| < \frac{\varepsilon |g(c)|^2}{2}$$

whenever $x \in D$ and $|x - c| < \delta_2$. Let $\delta = \min(\delta_1, \delta_2)$. Then, for every $\varepsilon > 0$, if $x \in D$ and $|x - c| < \delta$, it follows that

$$\begin{aligned} \left| \frac{1}{g(x)} - \frac{1}{g(c)} \right| &= \frac{|g(x) - g(c)|}{|g(c)||g(x)|} \\ &< \frac{2\varepsilon |g(c)|^2}{2|g(c)|^2} \\ &= \varepsilon. \end{aligned}$$

□

Proof. (iv) Exercise. □

With this theorem, we are now able to immediately conclude more complicated functions, namely functions that are built from the addition, multiplication, division or scalar multiplication of simpler continuous functions at a point, are also continuous at that same point. This is of great use since we no longer need to check the ε-δ-definition of continuity for a given function at a point if it is simply built from these simpler ones. As always with mathematicians, the purpose of general theorems is to ensure less work is needed overall and to generally condense proofs. To demonstrate this theorem in action, consider the following example.

Example 4.2: Polynomials

Consider the function $p : \mathbb{R} \to \mathbb{R}$ given by

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

where $a_i \in \mathbb{R}$ for each $i \in \{0, 1, \ldots, n\}$ and $n \in \mathbb{N} \cup \{0\}$. These functions are called ***polynomials of order*** n. Since $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x$ is continuous for every $c \in \mathbb{R}$, it follows by Theorem 4.1 that p is continuous for every $c \in \mathbb{R}$.

When constructing new function from old, the operation of function composition was also considered. Like with the algebraic operations on functions, it can be shown that function composition also preserved the continuity of a function at a point, providing from additional conditions are satisfied. These are outlined in the following theorem.

Theorem 4.3: Composition of Continuous Functions

Let $f : D_1 \to \mathbb{R}$ and $g : D_2 \to \mathbb{R}$ with the ***range*** $f(D_1) := \{f(x) : x \in D_1\}$ a subset of D_2. Then if f is continuous at $c \in D_1$, and g is continuous at $f(c) \in f(D_1)$, the composition $g \circ f$ is continuous at $c \in D_1$.

Proof. For every $\varepsilon > 0$, there exists a $\delta_1 > 0$ such that

$$|g(x) - g(f(c))| < \varepsilon$$

whenever $x \in D_2$ and $|x - f(c)| < \delta_1$. Since $f(D_1) \subset D_2$ it follows that

$$|g(f(x)) - g(f(c))| < \varepsilon$$

whenever $f(x) \in D_2$ and $|f(x) - f(c)| < \delta_1$. Since f is continuous at c, it follows that there exists a $\delta > 0$ such that

$$|f(x) - f(c)| < \delta_1$$

whenever $x \in D_1$ and $|x - c| < \delta$. Hence, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|g(f(x)) - g(f(c))| < \varepsilon$$

whenever $x \in D_1$ and $|x - c| < \delta$. □

Like in the case of Theorem 4.1, the composition of continuous functions theorem can be utilised to condense the proof of the continuity of complicated functions at a point. Further, it can even be utilised to prove a function is continuous at every point in the domain by composing the function with the family of affine functions $f_a : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x + a$ for each $a \in \mathbb{R}$. An example of this is given as follows.

Example 4.4: Function Composition

The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = \sin(x)$ is continuous on $\mathbb{R}$.

Proof. We start by showing f is continuous at the point 0. For every $\varepsilon > 0$, let $\delta = \varepsilon$. Then since $|\sin(x)| \leq |x|$ for all $x \in \mathbb{R}$ it follows that

$$|\sin(x)| \leq |x| < \delta = \varepsilon,$$

whenever $|x| < \delta$. Similarly, the function $h : \mathbb{R} \to \mathbb{R}$ given by $h(x) = \cos(x)$ is continuous at the point 0. This follows since $|\cos(x) - 1| \leq |x|$ for all $x \in \mathbb{R}$. Hence, for every $\varepsilon > 0$, let $\delta = \varepsilon$, then

$$|\cos(x) - 1| \leq |x| < \delta = \varepsilon,$$

whenever $|x| < \delta$. Now the function $g_a : \mathbb{R} \to \mathbb{R}$ given by $g(x) = x - a$ is continuous at each $c \in \mathbb{R}$ and for every $a \in \mathbb{R}$ (*Exercise*). By Theorem 4.3, it follows that $f \circ g : \mathbb{R} \to \mathbb{R}$ given by $f(g(x))) = \sin(x - a)$ is continuous at

the point $x = a$, for each $a \in \mathbb{R}$. Since

$$\sin(x) = \sin(x - a)\cos(a) + \cos(x - a)\sin(a),$$

it follows by Theorem 4.1 that $\sin(x)$ is continuous for each $x \in \mathbb{R}$. □

So, to conclude this section, continuity of a function at a particular point can be proven directly from Definition 3.3, or by utilising Theorem 4.1 or Theorem 4.3, and the knowledge of the continuity of simple functions at specific points.

5 Discontinuous Functions

In the section on continuous functions, we have claimed that not all functions that contain no gaps are continuous. Given this, it is sensible to define a notion of a function to be not continuous at a point. It would seem sensible for this to be a definition of a function to be *discontinuous* at a point. However, this definition doesn't suffice in certain cases. In particular, consider the function $f : (0, \infty) \to \mathbb{R}$ given by $f(x) = 1/x$. Suppose we wish to know whether f is continuous at the point $c = 0$. Well, since $0 \notin (0, \infty)$ it follows that f is not defined at 0. Hence, it is not possible to ask whether f is continuous at the point 0 since it is simply not defined there. We could get around this by adding the point 0 to the domain of the function defining a new function $g : [0, \infty) \to \mathbb{R}$ given by

$$g(x) = \begin{cases} 1/x & : (0, \infty) \\ p & : x = 0 \end{cases},$$

with $p \in \mathbb{R}$. With this new function g, we can now ask if g is continuous at the point $c = 0$, since 0 is now in the domain of g. Well, since $f(x) = 1/x$ is unbounded, it means that for every $\delta > 0$, there exists an $x \in (0, \infty)$ with $|x| < \delta$ and

$$\left| \frac{1}{x} - p \right| \geq 1.$$

This means that g is not continuous at 0, and importantly this is true regardless of the value of $p \in \mathbb{R}$. So by adding 0 to the domain of f this allowed us to analyse the continuity at this point, but no matter how we extend f so its domain includes 0, it will always be discontinuous at this point. We ideally would like a definition of discontinuity that would allow us to say that a function like $f(x) = 1/x$ is discontinuous at $x = 0$, without extending the functions domain. In other words, given a general real function $f : (a, b) \to \mathbb{R}$, we would like our definition of discontinuity to encompass discontinuity at the points a and b, despite them not being in the set. In general, these boundaries a, b of any open interval (a, b) are a special type of point known as a *limit point*. These are formally defined below.

Definition 5.1: Limit Points and Closure

A point $c \in \mathbb{R}$ is a ***limit point*** of a set D if for every ε-neighbourhood $V_\varepsilon(c)$ of c there exists a point $x \in V_\varepsilon(c) \cap D$ with $x \neq c$. The ***closure*** of the set D

is the union of D and all its limit points, it is denoted by $\overline{D}$.

Based on the previous discussion, we would expect that an interval $(a, b) \subset \mathbb{R}$ has closure $[a, b]$. That is, the limit points of the set that are not in (a, b) are a and b themselves. An example proof of the limit points of a set of this type is given as follows.

Example 5.2: Limit Points

Let $D = (0, \infty)$. Then 0 is a limit point of D.

Proof. For every $\varepsilon > 0$, the set $V_\varepsilon(0) = (-\varepsilon, \varepsilon)$. It follows that $V_\varepsilon(0) \cap (0, \infty) = (0, \varepsilon) \subset (0, \infty)$. Hence $0 \in \overline{D}$. □

With this notion of limit point, we are now able to produce a definition of a discontinuous function that encompasses examples like $f(x) = 1/x$ as discussed above.

Definition 5.3: Discontinuous Function

A function $f : D \to \mathbb{R}$ is said to be ***discontinuous at*** $c \in \overline{D}$ if it is either not continuous at the point $c \in \overline{D}$ or $c \notin D$.

Let's verify that this definition encompasses the example $f(x) = 1/x$.

Example 5.4: Discontinuous Function

Let $f : (0, \infty) \to \mathbb{R}$ given by $f(x) = 1/x$. Then since 0 is a limit point of the set $(0, \infty)$ and $0 \notin (0, \infty)$, the function f is discontinuous at the point $x = 0$.

Now, the astute reader may quite rightly be critical of this definition of discontinuity for a function such as $f : (0, 1) \to \mathbb{R}$ with $f(x) = x$. Here, according to our definition of discontinuous we would say f is discontinuous at 0 and 1. This seems strange, as we would expect such a function to be continuous. In particular, unlike the function given by $1/x$, it is possible to extend the domain of this function to $[0, 1]$ so the function is now continuous at 0 and 1. Due to this, it is therefore taken implicitly that when one defines a continuous function, its domain is taken to be either of the form $[a, b]$, $(-\infty, a]$, $[b, \infty)$ or $\mathbb{R}$. In this sense, there are no limit points of the set that are not already in the domain. So, for the example $f(x) = x$, it would generally be defined on the domain $f : \mathbb{R} \to \mathbb{R}$ since this is the maximal domain for which the function is continuous.
In the opening sections, considerable time was spent discussing the distinctions between graphs with no gaps and continuous functions. In particular, an example of a function that had no gaps but was not continuous was the so called topological sine curve. In the following example, we demonstrate that this is indeed discontinuous at the point 0.

Example 5.5: Discontinuous Function

Let $f : [0,1] \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \sin(1/x) & : x \in (0,1] \\ 0 & : x = 0 \end{cases}$$

Then f is discontinuous at $c = 0$.

Proof. Let $\varepsilon = 1/2$. Then for every $\delta > 0$, there exists an $n \in \mathbb{N}$ such that

$$0 < \frac{2}{(2n-1)\pi} < \delta.$$

Since

$$\left|\sin\left(\frac{1}{2/((2n-1)\pi)}\right)\right| = \left|\sin\left(\frac{\pi(2n-1)}{2}\right)\right| = 1 \geq 1/2,$$

it follows that f is discontinuous at the point $c = 0$. □

6 Functional Limits

Like sequences, general real functions may have unbounded domains. It may be that a function is modelling the behaviour of a particular dynamical system. For example, if we think of the domain of the function as a time input, a function could describe the trajectory of a particle or even the value of a particular stock at each point in time. It would therefore be useful to know what the long term behaviour is of the function. Perhaps at time evolves, it ends up arbitrarily close to a particular value, perhaps it is unbounded or perhaps it approaches no value at all. However, unlike sequences, general real functions can be unbounded from below. So we could equally ask what the behaviour of the function was like a long time ago in the past. This could be useful in a situation where we want to understand how the initial conditions of a dynamical system effect its evolution. What we are building towards here is the notion of an infinite limit of a real function. This is defined in a very similar way to that of a sequence, with its only difference really being the domain on which the funtion is defined. It is given explicitly as follows.

Definition 6.1: Infinite Limits

Let $f : (a, \infty) \to \mathbb{R}$. Then f ***converges to*** $L \in \mathbb{R}$ ***as*** x ***goes to infinity*** if for every $\varepsilon > 0$, there exists an $M > a$ such that

$$|f(x) - L| < \varepsilon$$

for every $x > M$. We write this as $\lim_{x\to\infty}(f(x)) = L$.

Similarly, let $g : (-\infty, b) \to \mathbb{R}$. Then g ***converges to*** $L \in \mathbb{R}$ ***as*** x ***goes to negative infinity*** if for every $\varepsilon > 0$, there exists an $M < b$ such that

$$|g(x) - L| < \varepsilon$$

for every $x < M$. We write this as $\lim_{x\to-\infty}(g(x)) = L$.

Note that the key point for defining infinite limits for both sequences and functions relied on this notion of being arbitrarily close to a point. In both cases, the functions became arbitrarily close to some value in the co-domain as larger and larger values in the domain were input. However, unlike sequences, general real functions can take the form $f : I \setminus \{c\} \to \mathbb{R}$, where $c \in I$ and I is some interval of $\mathbb{R}$. By the completeness properties of $\mathbb{R}$, this means that we could get arbitrarily close to some number $c \in I$ without actually equalling it. We could then ask if the function also gets arbitrarily close to a value in its domain as we get arbitrarily close this value c. Essentially, we are arriving at the notion of a limit at a point.

Definition 6.2: Limits of Functions

Let $f : D \to \mathbb{R}$ and let $c \in \overline{D}$. Then f ***converges to*** $L \in \mathbb{R}$ ***as*** x ***approaches*** $c \in \overline{D}$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon$$

whenever $0 < |x - c| < \delta$ and $x \in D$. We write this as $\lim_{x\to c}(f(x)) = L$

Note an important point of this definition, namely it may be that the function is not defined at the point c, yet its limit could be. That is, limits are a slightly broader notion that continuity. To see this explicitly, consider the following example.

Example 6.3: Limits of Functions

Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by $f(x) = x\sin(1/x)$. Then $\lim_{x\to 0} f(x) = 0$.

Scratch Work. Let's start by manipulating the expression

$$\begin{aligned} |f(x) - L| &= \left|x\sin\left(\frac{1}{x}\right) - 0\right| \\ &\leq |x|. \end{aligned}$$

So, by choosing $\delta = \varepsilon$, we should get our desired result.

Proof. For every $\varepsilon > 0$, define $\delta = \varepsilon$. Then, if $|x| < \delta$ and $x \in \mathbb{R} \setminus \{0\}$, it follows that

$$\begin{aligned} |f(x) - 0| &= \left|x\sin\left(\frac{1}{x}\right) - 0\right| \\ &\leq |x| \\ &< \delta \\ &= \varepsilon. \end{aligned}$$

□

The astute reader may notice the similarity between the definition of a limit at a point and that of continuity. In particular, the only difference is really around

if the point c is in the domain or not. Given this, the definition of continuity is often given in terms of limits due to the following theorem.

Theorem 6.4: Limits and Continuity

Let $f : D \to \mathbb{R}$. The function is continuous at the point $c \in D$ if and only if $\lim_{x\to c}(f(x)) = f(c)$.

Proof. Exercise □

Similar to continuous functions, if the limits of functions at a point exist, they are well behaved under the algebraic operations of $\mathbb{R}$. This is outlined in the following theorem.

Theorem 6.5: Algebraic Properties of Limits

Let $f : D \to \mathbb{R}$ and $g : D \to \mathbb{R}$ be real functions such that $\lim_{x\to c}(f(x)) = L$ and $\lim_{x\to} g(x) = M$ for some $c \in \overline{D}$. Then,

(i) $\lim_{x\to c}(\lambda f(x)) = \lambda L$, for all $\lambda \in \mathbb{R}$,

(ii) $\lim_{x\to c}(f(x) + g(x)) = L + M$,

(iii) $\lim_{x\to c}(f(x)g(x)) = LM$, and

(iv) $\lim_{x\to c}\left(\frac{f(x)}{g(x)}\right) = \frac{L}{M}$, providing $g(x) \neq 0$ for all $x \in D$ and $M \neq 0$.

Proof. Exercise □

Consider the function $f : [0, 2] \to \mathbb{R}$ given by

$$f(x) = \begin{cases} x & : x \in [0, 1] \\ 1/2 & : (0, 2] \end{cases}.$$

This was given in an earlier section as an example of a function with gaps. Now, even though this function is very different on either side of the point $x = 1$, it is possible to define notions of limits from both sides. More specifically, we could restrict ourselves to get arbitrarily close to the point 1 by only approaching from the left and equally we could restrict ourselves to get arbitrarily close to the point 1 by only approaching from the right. That is, we can break up our notion of a limit at a point to two one sided limits. This gives the following definition.

Definition 6.6: One-sided Limits

Let $f : D \to \mathbb{R}$ and $c \in \overline{D}$.Then f ***converges to*** $L \in \mathbb{R}$ ***as*** x ***approaches*** $c \in \overline{D}$ ***from the right*** if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon$$

whenever $c < x < c + \delta$ and $x \in D$. We write this as $\lim_{x\to c^+}(f(x)) = L$

Similarly, f ***converges to*** $L \in \mathbb{R}$ ***as*** x ***approaches*** $c \in \overline{D}$ ***from the left*** if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon$$

whenever $c - \delta < x < c$ and $x \in D$. We write this as $\lim_{x \to c^-}(f(x)) = L$.

One sided limits can be useful for proving a function is discontinuous at a point. In particular, if we are able to show that the one sided limits from either side of a given point exist, but are not equal, then it follows that the function is discontinuous at that point.

Example 6.7: One-sided Limit

Let $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \sqrt{x} & : x \geq 0 \\ -x - 1 & : x < 0 \end{cases}.$$

Then $\lim_{x \to 0^+}(f(x)) = 0$ and $\lim_{x \to 0^-}(f(x)) = -1$ and hence f is discontinuous at $c = 0$.

Scratch Work. We break these into two parts, namely when $x \geq 0$ and $x < 0$. When $x \geq 0$ we have that

$$|f(x) - 0| = |\sqrt{x} - 0| = \sqrt{|x|}.$$

Thus, if we let $\delta = \epsilon^2$ we will get our desired result for the limit on the right. Now, if $x < 0$, we have that

$$|f(x) + 1| = |x|.$$

Thus, if $\delta = \varepsilon$ we will get our desired result for the limit on the left. We are now able to construct our proof.

Proof. Given $\varepsilon > 0$, let $\delta_1 = \varepsilon^2$ and $\delta_2 = \varepsilon$. Then, if $0 < x < \delta_1$, it follows that

$$|f(x) - 0| = |\sqrt{x}| < \sqrt{\delta_1} = \varepsilon.$$

Similarly, if $-\delta_2 < x < 0$ it follows that

$$|f(x) - 1| = |x| < \delta_2 = \varepsilon.$$

Since $\lim_{x \to 0^+}(f(x)) = 0 \neq -1 = \lim_{x \to 0^-}(f(x))$, it follows that the function is discontinuous at $c = 0$. □

7 Sequential Continuity

The initial notion of a limit was introduced for sequences and was then extended to more general real functions. In fact, it is possible to relate the two through the fact that any limit point is necessary the value of a particular convergent sequences. This theorem is outlined explicitly as follows.

Theorem 7.1: Sequences and Limit Points

A point $x \in \mathbb{R}$ is a limit point of a set $D \subset \mathbb{R}$ if and only if there exists a sequence (a_n) with $\lim(a_n) = x$ and $a_n \in D \setminus \{x\}$ for each $n \in \mathbb{N}$.

Proof. *Exercise.* □

Utilising this theorem, we can relate the notion of a limit of a function to that of a sequence. We will see that this equivalence is particularly useful for proving the discontinuity of particular functions.

Theorem 7.2: Sequential Criterion for Functional Limits

Let $f : D \to \mathbb{R}$ and $c \in \overline{D}$. Then $\lim_{x\to c}(f(x)) = L$ if and only if every sequence (a_n) with $\lim(a_n) = c$ and $a_n \in D \setminus \{c\}$ satisfies $\lim(f(a_n)) = L$.

Proof. ($\Rightarrow$) Suppose $\lim_{x\to c}(f(x)) = L$. Then for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon$$

whenever $0 < |x - c| < \delta$ and $x \in D$. Then a sequence (a_n), satisfying the required properties exists by Theorem 7.1, and there exists an $N \in \mathbb{N}$ such that

$$0 < |a_n - c| < \delta$$

for all $n > N$. Hence,

$$|f(a_n) - L| < \varepsilon$$

for all $n > N$.
($\Leftarrow$) Suppose towards a contradiction that there exists a sequence (a_n) with the required properties and $\lim(f(a_n)) = L$, as well as $\lim_{x\to c}(f(x)) \neq L$. Then there exists a $\varepsilon > 0$ such that for every $\delta > 0$, there exist $x \in D$ with $0 < |x - c| < \delta$ and $|f(x) - L| \geq \varepsilon$. Hence, for every $n \in \mathbb{N}$, there exists an $a_N \in D \setminus \{c\}$ with $0 < |a_N - c| < 1/n$ and $|f(a_n) - L| \geq \varepsilon$. It follows that $\lim(a_n) = c$, but then $|f(a_n) - L| \geq \varepsilon$ for every $n > N$. This is a contradiction. □

Note that this theorem is rather useless for proving the limit of a function is a particular value. The primary reason for this is that we would need to check the conditions in Theorem 7.2 for every single sequence. So why bother to prove this at all? Well actually, it turns out that it is the contrapositive of this theorem that is useful and be used as a test for discontinuity.

Corollary 7.3: Divergence Criterion for Functional Limits

Let $f : D \to \mathbb{R}$ and $c \in \overline{D}$. Then $\lim_{x\to c}(f(x))$ does not exist if there exists two distinct sequences (a_n) and (b_n), with $\lim(a_n) = \lim(b_n) = c$ and $a_n, b_n \in D \setminus \{c\}$ for each $n \in \mathbb{N}$, such that $\lim(f(a_n)) \neq \lim(f(b_n))$.

This corollary is extremely useful for proving the discontinuity of functions at particular point. In fact, it is worth going back to the discontinuous function section and seeing if you can prove the discontinuity of those example using this corollary, as opposed to directly from the ε-$\delta-$definition. Further though, we can utilise this corollary to prove that the *Dirichlet function* is discontinuous at every single point in its domain.

Example 7.4: Dirichlet Function

Recall that the Dirichlet function $f : \mathbb{R} \to \mathbb{R}$ is given by

$$f(x) = \begin{cases} 1 : x \in \mathbb{Q} \\ 0 : x \notin \mathbb{Q} \end{cases}.$$

Then f is not continuous at any point $c \in \mathbb{R}$.

Proof. Let $c \in \mathbb{R}$. Then there exists $a, b \in \mathbb{R}$ with $c \in (a, b)$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists an $a_1 \in \mathbb{Q}$ with $a_1 \in (a, c)$. Similarly, there exists $a_2 \in \mathbb{Q}$ with $a_2 \in (a_1, c)$. This defines a monotone increasing sequence (a_n) with $a_n \in \mathbb{Q}$ for each $n \in \mathbb{N}$ and $\lim(a_n) = c$. Additionally, since $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$, there exists a $b_1 \in \mathbb{R} \setminus \mathbb{Q}$ such that $b_1 \in (c, b)$. Using a similar argument as before, there exists a monotone decreasing sequence (b_n) with $b_n \in \mathbb{R} \setminus \mathbb{Q}$ for each $n \in \mathbb{N}$ and $\lim(b_n) = c$. Now, $f(a_n) = 1$ and $f(b_n) = 0$ for each $n \in \mathbb{N}$. Since $\lim(f(a_n)) = 1 \neq 0 = \lim(f(b_n))$ it follows by Corollary 7.3 that $\lim_{x\to c}(f(x))$ does not exist for any $c \in \mathbb{R}$. Thus, f is not continuous at any point $c \in \mathbb{R}$. □

8 Intermediate and Extreme Value Theorem

So far, we have seen that this definition of continuity resolves one of the problems that emerged by thinking of continuous functions as ones with no gaps. Namely, the sum of two continuous functions is also continuous. This was not the case for Darboux functions. There was a further issue though with Darboux functions. This was that Darboux functions defined on closed intervals, such at $[a, b]$, could be unbounded. We claimed that whatever our definition of continuity, that a continuous function on closed intervals should be bounded. The following theorem demonstrates that this is indeed the case for our definition of continuity and thus this definition resolves all the problems that occurred for Darboux functions.

Theorem 8.1: Boundedness Theorem

Let $f : [a, b] \to \mathbb{R}$ be a continuous function on $[a, b]$. Then the function is bounded.

Proof. Assume towards a contradiction that f is unbounded. Assume it is not bounded from above. Then it follows that there exists a sequence (a_n) with $a_n \in [a, b]$ and $f(a_n) > n$ for each $n \in \mathbb{N}$. Since $a_n \in [a, b]$ for each $n \in \mathbb{N}$, it follows by the Bolzano-Weierstrass theorem that there exists a subsequence $a_{n_k} \in [a, b]$ with $\lim(a_{n_k}) = L \in [a, b]$. By Theorem 7.2, it follows that $\lim(f(a_{n_k})) = f(L)$ and hence the sequence $(f(a_{n_k}))$ is bounded. However, by the construction of (a_n) it follows that

$$f(a_{n_k}) > n_k \geq k$$

for all $k \in \mathbb{N}$. That is, $(f(a_{n_k}))$ is unbounded. This is a contradiction. A similar argument can be produced by assuming the function is not bounded from below. This is left as an exercise. □

Whilst this theorem proves that our definition of continuity resolves the issue of unbounded functions on closed sets it actually goes one step further. Namely, a function is not only bounded, but it actually attains its bounded. This theorem is given a special name and is known as the *extreme value theorem.*

Theorem 8.2: Extreme Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be a continuous function on $[a, b]$. Then the function attains its bounds.

Proof. By Theorem 8.2, the function f is bounded. Let $S = \{f(x) : x \in [a, b]\}$. Since it is bounded, if follows by the completeness axiom that $\sup(S) = \alpha$ exists. It follows then that for every $n \in \mathbb{N}$, there exists a $a_n \in [a, b]$ such that

$$\alpha - \frac{1}{n} < f(a_n) \leq \alpha.$$

Hence, by the squeeze theorem, $\lim(f(a_n)) = \alpha$. Now, by the Bolzano-Weierstrass theorem, there exists a subsequence (a_{n_k}) with $\lim(a_{n_k}) = c$. Since $[a, b]$ is closed, $c \in [a, b]$ (*Exercise*). As f is continuous on $[a, b]$, it follows that $\lim(f(a_{n_k})) = f(c) = \alpha$. A similar argument can be produced to show that there exists a point $d \in [a, b]$ such that $f(d) = \inf(S)$ (*Exercise*). □

The reason this theorem has a special name is two fold. Firstly, it is utilised in many other areas of mathematics to prove much deeper results. Additionally, as hinted from its proof, it turns out that the extreme value theorem, like the Bolzano-Weierstrass theorem, is equivalent to the least upper bound property of the reals. That is, we could have defined the real numbers and proven all the theorems of real analysis by instead taking the extreme value theorem as an axiom. This is not at all clear and the proof of this is left to the most dedicated reader.

We now arrive at one of the most important theorems in real analysis, and arguably all of mathematics. This theorem is known as the *intermediate value theorem*. This theorem is outlined as follows.

Theorem 8.3: Intermediate Value Theorem

Let $f : D \to \mathbb{R}$ be a continuous functions on D. Then for every $[a, b] \subset D$, and $\alpha \in \mathbb{R}$, with $f(a) < \alpha < f(b)$ or $f(b) < \alpha < f(a)$, there exists a $c \in (a, b)$ such that $f(c) = \alpha$.

Proof. Let $[a, b] \subset D$ with $\alpha \in \mathbb{R}$ and $\alpha \in (f(a), f(b))$. Then the set

$$S = \{x \in [a, b] : f(x) \leq \alpha\}$$

is bounded above and thus by the completeness axiom, the supremum $\sup(S) = c \in \mathbb{R}$ exists. Further, since $[a, b]$ is closed, it follows that $c \in S$. Suppose first that $c \in (a, b)$. Then since $c = \sup(S)$, it follows that there exists $a_n \in S$ such that

$$c - \frac{1}{n} < a_n \leq c$$

for each $n \in \mathbb{N}$. Defining another sequence (b_n) by

$$b_n = c + \frac{b - c}{n},$$

it follows that

$$c - \frac{1}{n} < a_n \leq c \leq b_n \leq c + \frac{b - c}{n}.$$

By the squeeze theorem, $\lim(a_n) = \lim(b_n) = c$. Hence, by Theorem 7.2, it follows that $f(c) = \lim(f(a_n)) \leq \alpha$ and $f(c) = \lim(f(b_n)) \geq \alpha$ and thus $f(c) = \alpha$. Now suppose that $b = c$. In this case, it follows that there exists a sequence (a_n) with $\lim(a_n) = c$ and $a_n \in S$ for each $n \in \mathbb{N}$. Again, by the Theorem 7.2, it follows that $f(c) = \lim(f(a_n)) \leq \alpha < f(c)$. Hence, $b \neq c$. Similarly, if $a = c$, there exists a sequence (a_n) with $a_n \in [a, b] \setminus S$ for each $n \in \mathbb{N}$ and $\lim(a_n) = c$. Then, $f(c) < \alpha \leq \lim(f(a_n)) = f(c)$. So, $c \neq a$. □

Now, the intermediate value theorem (IVT) may not seem that profound, but in fact it was this theorem that led to the definition of continuity in the first place and in particular, is what highlighted the distinction between a graph with no gaps and continuity. The latter half of the theorem is precisely defining a Darboux function and therefore it was this theorem that specifically highlighted that Darboux functions and continuous functions are entirely different concepts. This is also why Darboux functions are sometimes referred to as *intermediate value theorem functions*, or *IVT functions*. Whilst this result was known before Bolzano proved it in 1817, it was taken to be vacuously true since all functions automatically satisfied this property, according to the understanding of function/continuity at the time. It was Bolzano's genius to isolate the concept of continuity from that of the IVT theorem. The title of Bolzano's original paper roughly translates to the following.

Purely analytic proof of the theorem that between any two values that yield results of opposite sign there will be at least one real root of the equation.

So it was the case that Bolzano only proved the IVT to prove a more specific theorem, namely that all cubics have at least one real root. In fact, if it wasn't for this theorem the concept of complex numbers may not have been developed until much later. That isn't to say that Bolzano's paper led to the development of complex numbers, after all, these were developed centuries before Bolzano was even born. But, this result that was taken to be true for centuries, and that led to the development of complex numbers, was finally proven on a rigours footing by Bolzano. The proof of this statement is outlined here, and is beautifully simple, as all great mathematics proofs should be.

Example 8.4: Roots of Cubics

Let $p : \mathbb{R} \to \mathbb{R}$ given by

$$p(x) = a_3x^3 + a_2x^2 + a_1x + a_0$$

where $a_i \in \mathbb{R}$ for each $i \in \{0, 1, 2, 3\}$. Then p has at least one real root.

Proof. Since $\lim_{x\to\infty}(p(x)) = \infty$ and $\lim_{x\to-\infty}(p(x)) = -\infty$, it follows that there exists an $a \in \mathbb{R}$ and $b \in \mathbb{R}$ such that $p(x) < 0$ for $x \leq a$ and $p(x) > 0$ for $x \geq b$. By Example 4.2, the function $p(x)$ is continuous on $\mathbb{R}$. It follows that $p|_{[a,b]} : [a.b] \to \mathbb{R}$ is continuous on $[a, b]$. Then by Theorem 8.3, it follows that there exists a $c \in (a, b)$ such that $p(c) = 0$. That is, p has at least one real root. □

It is also worth mentioning that like the extreme value theorem, the intermediate value theorem is equivalent to the least upper bound property of $\mathbb{R}$. In turn, this means that the exptreme value theorem and the intermediate value theorem are equivalent. This proof again is left as an exercise to the most dedicated reader.

This concludes the section on functions and continuity. In the final chapter we will address the notion of whether a continuous function can be approximating by a linear function at each point. This question naturally leads us to the notion of *differentiability* and in turn, the concept of a *Taylor series*. We will explore these notions further in the final chapter.